



Figure 7: Overall call flow

environment. All the configuration files are placed in `/etc/asterisk/` with an extension of `.conf`, ex: `sip.conf`, `extension.conf`.

`Sip.conf` and `extension.conf` are two key files essential for establishing communication links between the nodes. You need to define the users in the `sip.conf` and its associated extensions in `extension.conf`.

Sip.conf

It is located in `/etc/asterisk/sip.conf` on the Asterisk server. Define the users here for Asterisk to authenticate, as follows:

```
[Rakesh]
type=friend ;
host=dynamic ;
secret=***** ;
```

With the above configuration in `sip.conf`, you can register the user, 'Rakesh'.

Extension.conf

This is another configuration file located in `/etc/asterisk/extensions.conf`. In it you need to define the extension numbers corresponding to the usernames in the `sip.conf`. The usernames will be mapped to the extensions and a communication link will be established at run time when the call trigger is raised. Asterisk provides flexibility to name alphanumeric extensions.

```
exten => 1000,1,verbose(Extn 1000 calling)
exten => 1000,2,Dial(SIP/Rakesh,30)
exten => 1000,3,Hangup()
```

With the above configuration, 'Rakesh' has been mapped to extension 1000.

Call flow with demos

Softphone to softphone

Table 2 shows the list of softphones running with their corresponding usernames and extensions defined in the configuration. Figure 2 illustrates the call flow between 1000 and 1006 extensions. When user 'Linux-1' dials 1006 from his softphone, 'Twinkle' generates a call event in Asterisk, as indicated. Asterisk then validates the extension 1006 in the configuration files and the call event is triggered.

Cross-platform communication—a demo

Table 1 gives the configuration of the machine in which Asterisk Server is installed.

Laptop model	Dell Inspiron 1510
Processor	AMD Sempron 3200+
RAM	2 GB
Hard disk	160 GB
Static IP	192.168.1.6
Others	Headphones and Mic

Table 1

Asterisk Server will now be running on the machine. All machines listed in Table 2 are connected to a LAN using a wireless router. The softphones are connected to the Asterisk server using IP address 192.168.1.6 from their respective workstations.

Machine	Operating system	Soft-phone	Users	Extension
AMD	Ubuntu 11.0	Twinkle	user-X	709905
Linux - Virtual	Ubuntu 11.0	Twinkle	user-Y	709903
Intel	Windows 7	Adore	Windows	709906
Spice Mi-270	Android 2.2	3com	Android	709908

Table 2

Demo – softphone to softphone

Open the 'Twinkle' softphone and click on the editor button to create a user profile. Enter the user name specified in `sip.conf`.

Log in to the softphone 'Twinkle' with user-X and user-Y in two separate sessions/machines.

The screen shot in Figure 3 shows the user-X extension (709905) on the left and user-Y (709903) on the right. As the user-X dials extension 709903, an incoming call trigger is raised on user-Y's dialogue box, showing the caller's ID. The call rings for 30 seconds.

Softphone to Gtalk

Observe the communication links on Asterisk and you